

Using the California Housing dataset because I have already used it in Week 9.

```
from sklearn.datasets import fetch_california_housing
import pandas as pd

housing = fetch_california_housing(as_frame=True)

df = housing.frame

df.head()
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
!pip install pycaret
```



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Downloading wurlitzer-3.1.1-py3-none-any.whl (8.6 kB)
Downloading choreographer-1.3.0-py3-none-any.whl (52 kB)
52.6/52.6 kB 3.7 MB/s eta 0:00:00
Downloading dash-4.4.1-py3-none-any.whl (8.9 MB)
8.9/8.9 MB 98.0 MB/s eta 0:00:00
Downloading jedi-0.20.0-py2.py3-none-any.whl (4.9 MB)
4.9/4.9 MB 97.9 MB/s eta 0:00:00
Downloading logistro-2.0.1-py3-none-any.whl (8.6 kB)
Downloading scikit_base-0.7.8-py3-none-any.whl (130 kB)
130.1/130.1 kB 10.9 MB/s eta 0:00:00
Downloading tsdownsample-0.1.5.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.3
1.3/1.3 MB 56.8 MB/s eta 0:00:00
Downloading janus-2.0.0-py3-none-any.whl (12 kB)
Downloading comm-0.2.3-py3-none-any.whl (7.3 kB)
Downloading retrying-1.4.2-py3-none-any.whl (10 kB)
Installing collected packages: wurlitzer, scikit-base, schemdraw, retrying, numpy, logistro, jobl
Attempting uninstall: numpy
  Found existing installation: numpy 2.0.2
  Uninstalling numpy-2.0.2:
    Successfully uninstalled numpy-2.0.2
Attempting uninstall: joblib
  Found existing installation: joblib 1.5.3
  Uninstalling joblib-1.5.3:
    Successfully uninstalled joblib-1.5.3
Attempting uninstall: scipy
  Found existing installation: scipy 1.16.3
  Uninstalling scipy-1.16.3:
    Successfully uninstalled scipy-1.16.3
Attempting uninstall: pandas
  Found existing installation: pandas 2.2.2
  Uninstalling pandas-2.2.2:
    Successfully uninstalled pandas-2.2.2
Attempting uninstall: scikit-learn
  Found existing installation: scikit-learn 1.6.1
  Uninstalling scikit-learn-1.6.1:
    Successfully uninstalled scikit-learn-1.6.1
Attempting uninstall: matplotlib
  Found existing installation: matplotlib 3.10.0
  Uninstalling matplotlib-3.10.0:
    Successfully uninstalled matplotlib-3.10.0
ERROR: pip's dependency resolver does not currently take into account all the packages that are i
google-colab 1.0.0 requires pandas==2.2.2, but you have pandas 2.1.4 which is incompatible.
hdbscan 0.8.44 requires scikit-learn>=1.6, but you have scikit-learn 1.4.2 which is incompatible.
xarray-einstats 0.10.0 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
xarray-einstats 0.10.0 requires scipy>=1.13, but you have scipy 1.11.4 which is incompatible.
xarray 2025.12.0 requires pandas>=2.2, but you have pandas 2.1.4 which is incompatible.
libpysal 4.14.1 requires scipy>=1.12.0, but you have scipy 1.11.4 which is incompatible.
jaxlib 0.7.2 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
jaxlib 0.7.2 requires scipy>=1.13, but you have scipy 1.11.4 which is incompatible.
tiffle 2026.4.11 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
spopt 0.7.0 requires scipy>=1.12.0, but you have scipy 1.11.4 which is incompatible.
tobler 0.14.0 requires joblib>=1.4, but you have joblib 1.3.2 which is incompatible.
tobler 0.14.0 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
tobler 0.14.0 requires pandas>=2.2, but you have pandas 2.1.4 which is incompatible.
tobler 0.14.0 requires scipy>=1.13, but you have scipy 1.11.4 which is incompatible.

```

```

from pycaret.regression import *

setup(
    data=df,
    target="MedHouseVal",
    session_id=42,
    verbose=False
)

best_model = compare_models()

```

```

shap 0.52.0 requires numpy>=2, but you have numpy 1.26.4 which is incompatible.
access 1.1.10.post3 requires scipy>=1.14.1, but you have scipy 1.11.4 which is incompatible.
arviz 0.22.0 requires matplotlib>=3.8, but you have matplotlib 3.7.5 which is incompatible.
mizani 0.13.5 requires pandas>=2.0, but you have pandas 2.1.4 which is incompatible.
rasterio 1.5.0 requires numpy>=2, but you have numpy 1.26.4 which is incompatible.
mapclassify 2.10.0 requires scipy>=1.12, but you have scipy 1.11.4 which is incompatible.
opencv_contrib_python 4.13.0.92 requires numpy>=2; python_version >= "3.9", but you have numpy 1.
jax 0.7.2 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
jax 0.7.2 requires scipy>=1.13, but you have scipy 1.11.4 which is incompatible.
spint 1.1.0 requires scipy>=1.12, but you have scipy 1.11.4 which is incompatible.
/usr/local/lib/python3.12/dist-packages/pycaret/_init_.py in <module>
pytensor 2.38.3 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
Successfully installed category-encoders-2.7.0 choreographer-1.3.0 comm-0.2.3 dash-4.4.1 janus-2.
WARNING: The following packages were previously imported in this runtime:
[joblib,matplotlib,mpl_toolkits,numpy,pandas,scipy,sklearn]
Pycaret only supports python 3.9, 3.10, 3.11. Your actual Python version: ",
You must restart the runtime in order to use newly installed versions.
sys.version_info,

```

RESTART SESSION

```

RuntimeError: ('Pycaret only supports python 3.9, 3.10, 3.11. Your actual Python version: ',
sys.version_info(major=3, minor=12, micro=13, releaselevel='final', serial=0), 'Please DOWNGRADE
your Python version.')

```

Next steps: [Explain error](#)

!pip install autogluon

```
Found existing installation: pyarrow 18.1.0
Uninstalling pyarrow-18.1.0:
  Successfully uninstalled pyarrow-18.1.0
Attempting uninstall: nltk
Found existing installation: nltk 3.9.1
Uninstalling nltk-3.9.1:
  Successfully uninstalled nltk-3.9.1
Attempting uninstall: fastcore
Found existing installation: fastcore 2.0.5
Uninstalling fastcore-2.0.5:
  Successfully uninstalled fastcore-2.0.5
Attempting uninstall: jsonschema
Found existing installation: jsonschema 4.26.0
Uninstalling jsonschema-4.26.0:
  Successfully uninstalled jsonschema-4.26.0
Attempting uninstall: timm
Found existing installation: timm 1.0.28
Uninstalling timm-1.0.28:
  Successfully uninstalled timm-1.0.28
Attempting uninstall: python-fasthtml
Found existing installation: python-fasthtml 0.14.6
Uninstalling python-fasthtml-0.14.6:
  Successfully uninstalled python-fasthtml-0.14.6
Successfully installed edantic 0.2.0 fasthtml 0.21.0 flask 1.0.1 flask-login 1.0.1 flask-wtf 1.0.1 fastapi 0.109.0
```

```
from autogluon.tabular import TabularPredictor
```

```
!pip install tpot
```



```

Collecting tpot
  Downloading TPOT-1.1.0-py3-none-any.whl.metadata (1.9 kB)
Requirement already satisfied: numpy>=1.26.4 in /usr/local/lib/python3.12/dist-packages (from tpo
Requirement already satisfied: scipy>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from tpot
Collecting scikit-learn>=1.6 (from tpot)
  Downloading scikit_learn-1.9.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.meta
Collecting update-checker>=0.16 (from tpot)
  Downloading update_checker-1.0.0-py3-none-any.whl.metadata (2.9 kB)
Requirement already satisfied: tqdm>=4.36.1 in /usr/local/lib/python3.12/dist-packages (from tpot
Collecting stopit>=1.1.1 (from tpot)
  Downloading stopit-1.1.2.tar.gz (18 kB)
  Preparing metadata (setup.py) ... done
Collecting pandas>=2.2.0 (from tpot)
  Downloading pandas-3.0.5-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (
79.5/79.5 kB 2.3 MB/s eta 0:00:00
Requirement already satisfied: joblib>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from tpo
Requirement already satisfied: xgboost>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from tp
Requirement already satisfied: matplotlib>=3.6.2 in /usr/local/lib/python3.12/dist-packages (from
Collecting traitlets>=5.8.0 (from tpot)
  Downloading traitlets-5.16.1-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: lightgbm>=3.3.3 in /usr/local/lib/python3.12/dist-packages (from t
Requirement already satisfied: optuna>=3.0.5 in /usr/local/lib/python3.12/dist-packages (from tpo
Requirement already satisfied: networkx>=3.0 in /usr/local/lib/python3.12/dist-packages (from tpo
Requirement already satisfied: dask>=2024.4.2 in /usr/local/lib/python3.12/dist-packages (from tp
Collecting distributed>=2024.4.2 (from tpot)
  Downloading distributed-2026.7.1-py3-none-any.whl.metadata (3.4 kB)
Collecting dask-expr>=1.0.12 (from tpot)
  Downloading dask_expr-2.0.0-py3-none-any.whl.metadata (2.3 kB)
Collecting dask-jobqueue>=0.8.5 (from tpot)
  Downloading dask_jobqueue-0.9.0-py2.py3-none-any.whl.metadata (1.3 kB)
Collecting func-timeout>=4.3.5 (from tpot)
  Downloading func_timeout-4.3.5.tar.gz (44 kB)
44.3/44.3 kB 3.5 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Collecting configspace>=1.1.1 (from tpot)
  Downloading configspace-1.2.2-py3-none-any.whl.metadata (8.0 kB)
Collecting dill>=0.3.9 (from tpot)
  Downloading dill-0.4.1-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: seaborn>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from t
Requirement already satisfied: pyparsing in /usr/local/lib/python3.12/dist-packages (from config
Requirement already satisfied: typing_extensions in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: more_itertools in /usr/local/lib/python3.12/dist-packages (from co
Requirement already satisfied: click>=8.1 in /usr/local/lib/python3.12/dist-packages (from dask>=
Requirement already satisfied: cloudpickle>=3.0.0 in /usr/local/lib/python3.12/dist-packages (fro
Requirement already satisfied: fsspec>=2021.09.0 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from d
Requirement already satisfied: partd>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from dask
Requirement already satisfied: pyyaml>=5.4.1 in /usr/local/lib/python3.12/dist-packages (from das
Requirement already satisfied: toolz>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from das
Collecting dask>=2024.4.2 (from tpot)
  Downloading dask-2026.7.1-py3-none-any.whl.metadata (3.8 kB)
Requirement already satisfied: Jinja2>=2.10.3 in /usr/local/lib/python3.12/dist-packages (from di
Requirement already satisfied: locket>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from di
Requirement already satisfied: msgpack>=1.0.2 in /usr/local/lib/python3.12/dist-packages (from di
Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from di
Collecting sortedcontainers>=2.0.5 (from distributed>=2024.4.2->tpot)
  Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl.metadata (10 kB)
Collecting tblib>=3.2.0,!=3.2.1,>=1.6.0 (from distributed>=2024.4.2->tpot)
  Downloading tblib-3.2.2-py3-none-any.whl.metadata (27 kB)
Requirement already satisfied: tornado>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from di
Collecting zict>=3.0.0 (from distributed>=2024.4.2->tpot)
  Downloading zict-3.0.0-py2.py3-none-any.whl.metadata (899 bytes)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matp
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: kiwisolver>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: pillow>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from mat
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (f
Requirement already satisfied: alembic>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from op
Requirement already satisfied: colorlog in /usr/local/lib/python3.12/dist-packages (from optuna>=
Requirement already satisfied: sqlalchemy>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from
Collecting joblib>=1.1.1 (from tpot)
  Downloading joblib-1.5.3-py3-none-any.whl.metadata (5.5 kB)

```

Requirement already satisfied: narwhals>=2.0.1 in /usr/local/lib/python3.12/dist-packages (from s
 Requirement already satisfied: threadpoolctl>=3.5.0 in /usr/local/lib/python3.12/dist-packages (f
 Requirement already satisfied: nvidia-nccl-cu12 in /usr/local/lib/python3.12/dist-packages (from
 Requirement already satisfied: Mako in /usr/lib/python3/dist-packages (from alembic>=1.5.0->optun
 Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from j
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-d
 Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.12/dist-packages (from sqlal

Downloading TPOT-1.1.0-py3-none-any.whl (215 kB)

215.1/215.1 kB 7.3 MB/s eta 0:00:00

Downloading configspace-1.2.2-py3-none-any.whl (117 kB)

117.6/117.6 kB 9.1 MB/s eta 0:00:00

Downloading dask_expr-2.0.0-py3-none-any.whl (3.2 kB)

Downloading dask_jobqueue-0.9.0-py2.py3-none-any.whl (52 kB)

52.0/52.0 kB 4.0 MB/s eta 0:00:00

Downloading dill-0.4.1-py3-none-any.whl (120 kB)

120.0/120.0 kB 9.8 MB/s eta 0:00:00

Downloading distributed-2026.7.1-py3-none-any.whl (1.0 MB)

1.0/1.0 MB 19.7 MB/s eta 0:00:00

Downloading dask-2026.7.1-py3-none-any.whl (1.5 MB)

1.5/1.5 MB 49.9 MB/s eta 0:00:00

Downloading pandas-3.0.5-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (11.0 MB)

11.0/11.0 MB 92.9 MB/s eta 0:00:00

Downloading scikit_learn-1.9.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (9.1 M

9.1/9.1 MB 84.6 MB/s eta 0:00:00

Downloading joblib-1.5.3-py3-none-any.whl (309 kB)

309.1/309.1 kB 19.9 MB/s eta 0:00:00

Downloading traitlets-5.16.1-py3-none-any.whl (86 kB)

86.2/86.2 kB 6.9 MB/s eta 0:00:00

Downloading update_checker-1.0.0-py3-none-any.whl (10 kB)

Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl (29 kB)

Downloading tblib-3.2.2-py3-none-any.whl (12 kB)

Downloading zict-3.0.0-py2.py3-none-any.whl (43 kB)

43.3/43.3 kB 3.3 MB/s eta 0:00:00

Building wheels for collected packages: func-timeout, stopit

Building wheel for func-timeout (setup.py) ... done

Created wheel for func-timeout: filename=func_timeout-4.3.5-py3-none-any.whl size=15077 sha256=

Stored in directory: /root/.cache/pip/wheels/ed/94/4a/9ccd71f7284a92e4137b7dd0ce8540c40ea7c2689

Building wheel for stopit (setup.py) ... done

Created wheel for stopit: filename=stopit-1.1.2-py3-none-any.whl size=11938 sha256=260360db5245

Stored in directory: /root/.cache/pip/wheels/10/10/63/c3c98c9859d2aa59553536cc2ea005d3c9c39e214

Successfully built func-timeout stopit

Installing collected packages: stopit, sortedcontainers, func-timeout, zict, update-checker, trai

Attempting uninstall: traitlets

Found existing installation: traitlets 5.7.1

Uninstalling traitlets-5.7.1:

Successfully uninstalled traitlets-5.7.1

Attempting uninstall: joblib

Found existing installation: joblib 1.3.2

Uninstalling joblib-1.3.2:

Successfully uninstalled joblib-1.3.2

Attempting uninstall: dill

Found existing installation: dill 0.3.8

Uninstalling dill-0.3.8:

Successfully uninstalled dill-0.3.8

Attempting uninstall: scikit-learn

Found existing installation: scikit-learn 1.4.2

Uninstalling scikit-learn-1.4.2:

Successfully uninstalled scikit-learn-1.4.2

Attempting uninstall: pandas

Found existing installation: pandas 2.1.4

Uninstalling pandas-2.1.4:

Successfully uninstalled pandas-2.1.4

Attempting uninstall: dask

Found existing installation: dask 2026.7.0

Uninstalling dask-2026.7.0:

Successfully uninstalled dask-2026.7.0

ERROR: pip's dependency resolver does not currently take into account all the packages that are i
 autogluon-core 1.6.1 requires pandas<2.4.0,>=2.0.0, but you have pandas 3.0.5 which is incompatib
 gluonts 0.17.0 requires pandas<3,>=1.4, but you have pandas 3.0.5 which is incompatible.
 category-encoders 2.7.0 requires scikit-learn<1.6.0,>=1.0.0, but you have scikit-learn 1.9.0 whic
 sktime 0.26.0 requires pandas<2.2.0,>=1.1, but you have pandas 3.0.5 which is incompatible.
 sktime 0.26.0 requires scikit-learn<1.5.0,>=0.24, but you have scikit-learn 1.9.0 which is incomp
 autogluon-timeseries 1.6.1 requires pandas<2.4.0,>=2.0.0, but you have pandas 3.0.5 which is inco

```
from tpot import TPOTRegressor
```

```
!pip install h2o
```

```
import h2o
```

```
h2o.init()
```

You must restart the runtime in order to use newly installed versions.

Checking whether there is an H2O instance running at <http://localhost:54321>..... not found.

Attempting to start a local H2O server...

RESTART SESSION

Java Version: [0.024s][warning][os,container] Cgroup memory controller path at '/sys/fs/cgroup'

Starting server from /usr/local/lib/python3.12/dist-packages/h2o/backend/bin/h2o.jar

Ice root: /tmp/tmpaq8xrea0

JVM stdout: /tmp/tmpaq8xrea0/h2o_unknownUser_started_from_python.out

JVM stderr: /tmp/tmpaq8xrea0/h2o_unknownUser_started_from_python.err

Server is running at <http://127.0.0.1:54321>

Connecting to H2O server at <http://127.0.0.1:54321> ... successful.

H2O_cluster_uptime:	04 secs
H2O_cluster_timezone:	Etc/UTC
H2O_data_parsing_timezone:	UTC
H2O_cluster_version:	3.46.0.12
H2O_cluster_version_age:	4 days
H2O_cluster_name:	H2O_from_python_unknownUser_q7jmgz
H2O_cluster_total_nodes:	1
H2O_cluster_free_memory:	3.149 Gb
H2O_cluster_total_cores:	2
H2O_cluster_allowed_cores:	2
H2O_cluster_status:	locked, healthy
H2O_connection_url:	http://127.0.0.1:54321
H2O_connection_proxy:	{"http": null, "https": null, "colab_language_server": "/usr/colab/bin/language_service"}
H2O_internal_security:	False
Python_version:	3.12.13 final
H2O_telemetry_server:	disabled
H2O_telemetry_client:	disabled

```
+-----+
| You are running the community edition of H2O-3 OSS.
|
| For commercial use, H2O-3 Secure is now recommended.
| This includes production support, CVE fixes, multi-node scaling,
| model artifact extraction, and more.
| See h2o.ai/h2o-3/oss-vs-secure for additional details.
| Contact enterprise@h2o.ai to upgrade.
+-----+
```

